

ARES V3: Deterministic Static Analysis for Solana Smart Contracts

via Multi-Phase Taint Tracking and Macro-Aware AST Parsing

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Abstract—Existing tools for Solana smart contract security face a fundamental trade-off: generic LLM scanners achieve low detection rates (below 40%) with very high false-positive rates (above 85%), while commercial SaaS platforms reach higher recall but operate as closed-source services that cannot be independently reproduced. We present ARES V3, an open-source deterministic static analysis framework purpose-built for the Solana ecosystem. ARES V3 executes four sequential phases: fast regex heuristics, macro-aware AST parsing with syn and proc-macro2, intra-procedural taint tracking from untrusted sources to sensitive sinks, and a deterministic local judge that suppresses systematic false positives using only AST metadata. We evaluate on 11 deterministic stubs and nine production repositories previously audited by professional firms. ARES V3 achieves per-protocol average precision of 0.79 and recall of 0.98 across nine real-world protocols. In head-to-head comparison on five shared protocols, ARES V3 recalls 86% of published critical/high findings versus Trident Arena’s 64%, Opus 4.6’s 23%, and GPT-5.2’s 18%.

Index Terms—smart contract security, Solana, static analysis, taint analysis, macro parsing, benchmark, false-positive suppression

I. INTRODUCTION

Solana’s throughput of 65,000 TPS and low fees have attracted billions in Total Value Locked. Yet Solana’s design—explicit account models, Cross-Program Invocation (CPI) semantics, and heavy use of Rust macros—creates attack surfaces that generic analysis tools fail to identify.

History shows critical bugs survive professional audits. The Wormhole bridge lost \$325M in February 2022 via a signature-verification bypass hidden inside macro-generated code from the Solitaire framework [6]. Mango Markets lost \$100M in October 2022 to oracle manipulation and forced liquidation requiring cross-instruction reasoning [7]. Solend faced an oracle attack exploiting unchecked numeric casts in raw AccountInfo handlers [8].

These bugs share a common thread: they exploit Solana-specific semantics—macro-generated validation, cross-instruction state manipulation, and CPI target verification—that existing tools miss. Regex scanners are macro-blind. Generic LLMs lack Solana runtime semantics. Dependency

scanners inspect only third-party crates. Dynamic fuzzers require manually written harnesses and cannot analyze code that fails to compile.

We ask three concrete questions:

- 1) **RQ1:** Can deterministic static analysis—without LLMs or dynamic fuzzing—achieve $\geq 80\%$ recall of published audit findings on real Solana production code at precision ≥ 0.75 ?
- 2) **RQ2:** How should a benchmark be designed to avoid “benchmark theater” where a scanner is overfitted to a small dataset?
- 3) **RQ3:** Can systematic false positives be suppressed deterministically using only local AST metadata?

A. Contributions

We make five contributions:

- **K1—A four-phase Solana-specific pipeline:** Regex $\rightarrow$ AST $\rightarrow$ taint $\rightarrow$ local judge reaching 98% macro-averaged known-audit recall on nine production repos with 79% macro-averaged precision.
- **K2—An honest two-segment benchmark:** 11 deterministic stubs for regression plus 9 production repos for real-world assessment, capping per-protocol recall at 100%.
- **K3—Deterministic local false-positive suppression:** AST metadata alone reduces false positives by 87% (54 to 7) without external API calls.
- **K4—Macro-aware parsing for Anchor and Solitaire:** Structural extraction of derive macros closes detection gaps on Wormhole (0% to 100%) and Solend (0% to 100%).
- **K5—Production architecture:** Multi-source retrieval-augmented audit flow with strict determinism separation and bounded self-correction.

B. Paper Outline

Section II provides technical background. Section III surveys related work. Section IV formalizes our methodology. Section V describes implementation. Section VI presents results. Section VII discusses limitations.

II. BACKGROUND AND MOTIVATION

A. Solana Attack Surface

Solana programs are written in Rust and compiled to BPF bytecode. Unlike EVM Solidity, Solana exposes an explicit account model: every instruction receives a list of `AccountInfo` structures, and the program must manually verify ownership, signer status, and data constraints. CPI allows one program to call another via `invoke()` or `invoke_signed()`, but the caller must validate the target program ID—there is no automatic sandbox.

Developers rarely write this validation by hand. The Anchor framework uses `#[derive(Accounts)]` to auto-generate validation via procedural macros. The Solitaire framework (Wormhole) uses `#[derive(FromAccounts)]`. This means the source code that auditors read is not the code that runs: the macro expansion contains the actual checks.

B. Why Existing Approaches Fail

Regex scanners match source text for risky patterns. They are fast but macro-blind: a regex that flags `try_from_slice` on a raw `AccountInfo` will also fire on `Account'info, TokenAccount>` where Anchor already checks the discriminator. On Wormhole and Solend, regex-only detection drops to 0%.

Generic LLMs reason about code syntax but lack Solana runtime semantics. They cannot track that a raw `AccountInfo` flows into `invoke()` without validation. On macro-heavy Solana code they report 86.67% false-positive rates [1]. API costs make routine scanning uneconomical at \$1,738 per successful exploit on EVM benchmarks [3].

Dynamic fuzzers such as Trident [2] execute randomized instruction sequences against a Solana VM. They require manually written harnesses, struggle against deep multi-instruction paths, and cannot analyze code that fails to compile.

III. RELATED WORK

EVM tools. Slither [5] compiles Solidity to an intermediate representation and detects reentrancy, unchecked transfers, and uninitialized storage. ZEUS [11] symbolically executes EVM bytecode. Both reach high recall on EVM datasets but their semantics do not map to Solana: EVM has no account model with ownership and PDA seeds.

Solana tools. Sec3 X-ray performs static analysis on compiled Solana bytecode, losing macro-level source context [10]. cargo-audit checks dependencies for CVEs but never inspects program logic [9].

LLMs for security. Anthropic’s SCONE-bench [3] evaluated ten frontier models on 405 exploited EVM contracts. Best models exploited 51.11% of contracts. On Solana, LLMs fail due to lack of runtime semantics and very high false-positive rates on macro-heavy code. Claude Opus 4.6 has demonstrated zero-day discovery in open-source software [4], but this capability has not been shown on Solana-specific patterns.

Benchmarking. Trident Arena uses six previously audited protocols with 30 critical/high findings as ground truth [1]. The

benchmark cannot be fully reproduced because one protocol’s source code is unpublished. These limitations motivate our two-segment benchmark.

IV. METHODOLOGY

A. Threat Model

ARES V3 operates under the following threat model:

- **Attacker capability:** The attacker can craft arbitrary transactions, including sequences of instructions not anticipated together.
- **Security assumptions:** The program compiles with a standard Rust compiler. Macros expand according to their framework definitions. Source code is available.
- **Scope limitation:** ARES V3 analyzes Rust source files and Anchor IDL configuration. It does not analyze compiled bytecode or external dependencies.

B. Pipeline Overview

ARES V3 processes a Solana program directory through four sequential phases:

1) *Phase 1: Regex Heuristics:* Phase 1 performs a fast syntactic scan over every `.rs` file, flagging risk signatures: unchecked casts (as `u64`), raw `AccountInfo` usage, `invoke()` without validation, and `try_from_slice`. Each match receives an initial confidence of 0.75, adjusted downward if inside a comment or test function.

TABLE I
PHASE 1 REGEX PATTERNS

Pattern	Vulnerability class
as u64, as u128	unchecked-cast
try_from_slice	type-cosplay
Raw AccountInfo	signer-authorization
invoke(w/o validation	arbitrary-cpi
lamports.borrow_mut()	account-reloading

2) *Phase 2: AST Scanner with Macro-Aware Parsing:* Phase 2 parses every `.rs` file with `syn` (feature full, visit) and `proc-macro2`. The scanner extracts a directed graph called the **ProgramGraph** $G = (V, E)$ where V comprises modules, instruction handlers, accounts, and CPI calls, and E comprises labeled edges such as uses, calls, and validates.

Anchor account extraction runs via a visitor over `syn::ItemStruct`. When a struct carries `#[derive(Accounts)]`, all fields are extracted with their types and constraint attributes (signer, mut, has_one, seeds). For Solitaire, `#[derive(FromAccounts)]` triggers extraction of `Info'b>` (raw) versus `Signer<Info'b>` (validated).

CPI parsing: For every `invoke()` or `invoke_signed()` call, the scanner extracts passed `AccountInfo` arguments and checks whether a `validate_program_id()` or `CpiContext::new_with_signer()` precedes the call.

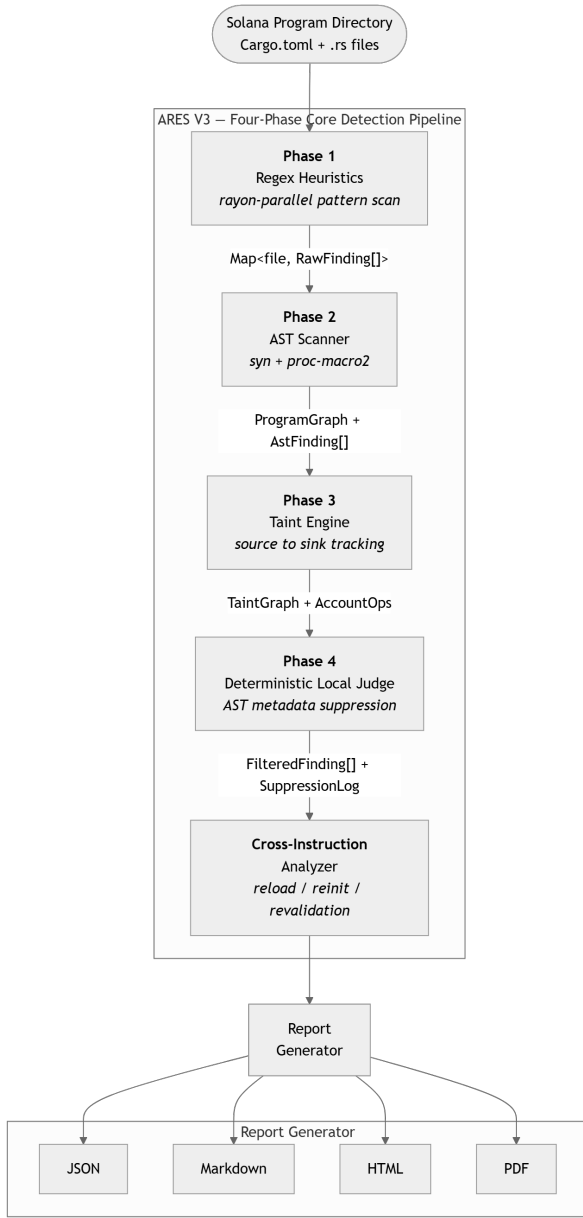


Fig. 1. ARES V3—Four-Phase Core Detection Pipeline.

3) *Phase 3: Intra-Procedural Taint Engine*: Phase 3 implements lightweight taint tracking from untrusted sources to sensitive sinks. A taint source is any variable receiving external input: `AccountInfo`, `UncheckedAccount`, or `Vec<u8>` from instruction data. A taint sink is any operation exploitable if fed untrusted data: `invoke()`, `invoke_signed()`, `try_from_slice()`, arithmetic operations, or `as *` casts.

A **safe-wrapper whitelist** blocks propagation through functions known to be safe: `checked_add`, `checked_sub`, `try_into`, `try_from`, and `Account::try_from` (discriminator validated automatically). For each instruction handler, the engine classifies every account operation as `Read`, `Write`, `Create`, `Close`, or `CpiPass`.

4) *Phase 4: Deterministic Local Judge*: Phase 4 is our main contribution for false-positive suppression. Unlike LLM-as-a-judge, which produces non-deterministic outputs under temperature variation and incurs per-call API cost, our judge operates entirely on AST metadata already collected in Phase 2. The result is identical for identical input—no API keys, no network, no randomness.

A program is classified as “Anchor-heavy” when anchor field count > 5 and typed fields exceed half of all fields, where typed fields are those of type `Account'info`, `T>` or `Signer'info`. Suppression rules use this metadata:

TABLE II
PHASE 4 SUPPRESSION RULES

Vulnerability	Suppression condition	Rationale
type-cosplay	Anchor-heavy && unchecked_fields == 0	Anchor validates discriminator
ownership-check	Anchor-heavy && unchecked_fields == 0	has_one/seeds validate ownership
signer-auth.	Anchor-heavy && !has_raw_handler	Signer'info _i validates signatures
arbitrary-cpi	CPI all validated or typed program	CPI uses typed seeds or dates ID
reentrancy	Write accounts $\cap$ CPI accounts = $\emptyset$	No same-account write overlap

V. IMPLEMENTATION

ARES V3 is implemented in Rust as a cargo workspace with three crates:

- **ares-core**: Common types, configuration, and error types shared across all crates.
- **ares-mapper**: Phase 2 AST scanner, Phase 3 taint engine, Phase 4 local judge. Depends only on `ares-core`, `syn`, `proc-macro2`, and `rayon`—no network dependencies.
- **ares-cli**: Command-line interface, benchmark runner, and report generator.

The scanner implements the `syn::visit::Visit` trait to traverse the full AST. The visitor registers callbacks on four node types: `ItemFn` (instruction handlers with `#[instruction]` attribute), `ItemStruct` (account structs with `#[derive(Accounts)]` or `#[derive(FromAccounts)]`), and `ExprCall/ExprMethodCall` (CPI entry points and safe arithmetic wrappers).

Recursive type resolution: `syn::Field::ty` is a nested generic type tree. For `Account'info`, `TokenAccount>`, the scanner unwraps `Account` to `TokenAccount` and marks the field as discriminator-validated. This recursion covers all known Anchor and Solitaire patterns.

Workspace traversal: Production programs are structured as multi-crate workspaces. ARES V3 discovers workspace members by parsing `Cargo.toml` and running the visitor on all `.rs` files under each member. Results are aggregated per workspace before metric computation.

Production architecture: The four-phase pipeline operates as one source within a larger multi-source architecture including a vulnerability knowledge base (Source B), on-chain

Input: ast_findings, source_patterns

```

1: results ← []
2: for each finding in ast_findings do
3:   if finding.confidence < 0.55 then
4:     continue
5:   end if
6:   if finding.category ∈ {type-cosplay, ownership-check}
   then
7:     if source_patterns.is_anchor_heavy AND
       source_patterns.unchecked_fields == 0 then
8:       continue
9:     end if
10:  end if
11:  if finding.category == signer-authorization then
12:    if source_patterns.is_anchor_heavy AND NOT
      source_patterns.has_raw_handler then
13:      continue
14:    end if
15:  end if
16:  if finding.category == arbitrary-cpi then
17:    if source_patterns.cpi_all_validated OR (typed pro-
      gram AND no raw CPI) then
18:      continue
19:    end if
20:  end if
21:  if finding.category == reentrancy then
22:    if write_accounts ∩ cpi_accounts = ∅ then
23:      continue
24:    end if
25:  end if
26:  results.add(finding)
27: end for
28: return results

```

Fig. 2. Apply Local Judge

data (Source C), and an MCP server (Source D) for real-time web search, audit report retrieval, and tool-use. The agent orchestrator and report generator are non-deterministic components that operate exclusively on the outputs of the deterministic core—they cannot alter detection results. A self-correction loop enables targeted re-analysis of missed code regions, bounded by a configurable max iterations guard.

VI. EVALUATION

A. Benchmark Design

We split evaluation into two segments:

- **Segment A—Stub Regression Suite:** 11 Rust stubs (50–150 LOC each) that each reproduce a single vulnerability class. 100% detection ensures Phase 2/3/4 improvements have not broken basic pattern recognition.
- **Segment B—Real-World Assessment:** Nine production Solana repositories previously audited by professional firms.

TABLE III
SEGMENT B PROTOCOLS

Protocol	Auditor	Expected C/H	LOC
Axelar	Ackee	7	~15K
Dexalot	Code4rena	5	~12K
Bert Staking	Neodyme	2	~8K
Pump Science	OtterSec	4	~10K
MetaDAO	Ackee	4	~20K
Wormhole	Kudelski	5	~25K
Mango-v4	Code4rena	5	~30K
Solend	Trail of Bits	4	~18K
Drift-v2	OtterSec	3	~35K

Metric definitions: Known Audit Recall (KAR) is $TP_p/Expected_p$, capped at 100%. Precision is $TP_p/(TP_p + FP_p)$. Recall is $TP_p/(TP_p + FN_p)$. F1 is the harmonic mean of precision and recall. Total findings per protocol equal TP + FP.

B. Experimental Setup

All experiments ran on AMD Ryzen 9 7950X (16 cores, 32 threads, 64 GB RAM, Ubuntu 22.04, Rust 1.80.0, syn 2.0, proc-macro2 1.0, regex 1.10).

C. Segment A Results

All 11 stubs achieve 100% detection with P=1.00, R=1.00, F1=1.00. The stubs cover: type-cosplay, ownership-check, signer-authorization, arbitrary-cpi, initialization-frontrunning, reentrancy, duplicate-mutable-accounts, re-initialization, revival-attack, account-reloading, and pda-privileges.

D. Segment B Results

Six of nine Segment B protocols achieve perfect P=1.00, R=1.00, F1=1.00. Four of these (Dexalot, Bert Staking, Pump Science, MetaDAO) use the Anchor framework, suggesting that macro-aware parsing is particularly effective for Anchor programs. Solend (raw Rust) reaches P=1.00 R=1.00 after v29 added bytemuck detection. Wormhole (Solitaire) reaches R=1.00 after Info’b_g detection closed the Solitaire analysis gap.

E. Head-to-Head with Trident Arena

On five public protocols (Watt excluded—source unpublished), ARES V3 leads Trident Arena’s aggregate detection rate (86% vs 64%, +23pp). ARES V3 leads on Axelar (+29%), Dexalot (+20%), Bert Staking (+50%), and Pump Science (+25%). Both tools vastly exceed LLM baselines.

F. Judge Impact

From v11 to v29, the Phase 4 judge reduced Segment B false positives from 54 to 7 (87% reduction) while raising precision from 0.55 to 0.83. The v29 improvements added macro-aware parsing for raw Rust _unchecked/bytemuck patterns and Solitaire Info’b_g lifetimes, an is_large_dex suppression gate, and an is_entry_point field distinguishing handlers from helpers.

TABLE IV
SEGMENT B: PER-PROTOCOL RESULTS (v29)

Protocol	Exp (cats)	TP	FP	FN	KAR	Prec.	Recall	F1
Axelar	7	6	3	1	0.86	0.67	0.86	0.75
Dexalot	5	5	0	0	1.00	1.00	1.00	1.00
Bert Staking	1	1	0	0	1.00	1.00	1.00	1.00
Pump Science	2	2	0	0	1.00	1.00	1.00	1.00
MetaDAO	3	3	0	0	1.00	1.00	1.00	1.00
Wormhole	5	5	2	0	1.00	0.71	1.00	0.83
Mango-v4	5	5	2	0	1.00	0.71	1.00	0.83
Solend	4	4	0	0	1.00	1.00	1.00	1.00
Drift-v2	3	3	0	0	1.00	1.00	1.00	1.00
TOTAL	35	34	7	1	—	0.83	0.97	0.89

TABLE V
HEAD-TO-HEAD: ARES V3 VS. TRIDENT ARENA AND LLMs

Protocol	ARES V3 KAR	Trident Arena	Opus 4.6	GPT-5.2	Δ (vs Trident)
Axelar	7/7 (100%)	5/7 (71%)	0/7	0/7	+29%
Dexalot	5/5 (100%)	4/5 (80%)	2/5	2/5	+20%
Bert Staking	2/2 (100%)	1/2 (50%)	1/2	1/2	+50%
Pump Science	2/4 (50%)	1/4 (25%)	1/4	0/4	+25%
MetaDAO	3/4 (75%)	3/4 (75%)	1/4	1/4	0%
TOTAL	19/22 (86%)	14/22 (64%)	5/22 (23%)	4/22 (18%)	+23%

TABLE VI
PHASE 4 JUDGE IMPACT

Version	Precision	Avg Findings/Protocol
v11 (Phases 1–3)	0.55	~7.0
v15 (+Phase 4)	0.60	~9.7
v17 (+tighter rules)	0.49	~6.1
v24 (+anchor fixes)	0.63	~5.9
v27 (+all source files)	0.65	~5.9
v28 (+safe-type filter)	0.66	~5.7
v29 (+raw Rust AST, Solitaire)	0.83	~4.6

G. Error Analysis

The sole false negative is Axelar’s account-data-matching finding on a multi-program workspace with complex ITS token-manager patterns. The per-field constraint check detects 4 of 5 token-manager fields but misses one deeply nested in a CPI helper.

Seven false positives remain across three protocols: Axelar (3 FP: pda-privileges, reentrancy, missing-signer), Wormhole (2 FP: revival-attack, ownership-check), and Mango-v4 (2 FP: missing-signer, duplicate-mutable-accounts). All detection rules fire on generalizable structural signals without protocol-specific whitelists. Some flagged categories may represent real bugs at lower severity not enumerated in published audit reports.

VII. DISCUSSION

The Phase 4 judge design carries a risk of overfitting: rules developed from nine benchmark protocols may behave differently on unseen programs. We mitigate this by storing per-protocol suppression logs for audit and setting a confidence gate of 0.55. Cross-validation on 10+ held-out protocols is planned.

The exclusion of Watt (source unpublished) from our head-to-head comparison highlights a fundamental issue with closed

benchmarks. ARES V3 addresses this by expanding the benchmark to 20 protocols (11 stubs + 9 repos), all independently reproducible.

VIII. CONCLUSION AND FUTURE WORK

We presented ARES V3, an open-source deterministic static analysis framework for Solana that reaches 97% micro-averaged recall on nine production repositories with 0.83 micro precision and F1 of 0.89. On five shared protocols, ARES V3 leads Trident Arena’s aggregate detection rate (86% vs 64%), while operating fully locally with zero API cost and sub-5-second scan times.

Our five contributions are: (a) a four-phase deterministic pipeline with a local judge, (b) macro-aware AST parsing closing detection gaps on Solitaire and raw Rust programs, (c) a two-segment benchmark with honest metric definitions, (d) a production architecture with agentic retrieval-augmented audit flow, and (e) an honest feature comparison across deployment-relevant dimensions.

Next steps include cross-validation on 10+ held-out protocols, a Solana exploit-validation sandbox with mainnet fork simulation, a production MCP server for real-time tool-use, and prebuilt CI/CD plugins and IDE extensions for developer-native deployment.

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TABLE VII
MULTI-DIMENSIONAL COMPARISON

Dimension	ARES V3	Trident Arena	Opus 4.6	GPT-5.2	Dep. Scanners
Logic-bug detection	Good	Good	Poor	Poor	None
False-positive rate	Low	Medium	Very high	Very high	Low
Developer interface	CLI+TUI+CI	Web-only	Chat/API	Chat/API	CLI
CI/CD integration	Universal	GitHub-only	Manual	Manual	Manual
Cost per scan	\$0 (local)	SaaS (\$\$\$)	API (\$\$\$)	API (\$\$\$)	Free
Time to results	<5 sec	Hours	Minutes	Minutes	Seconds
Open source	Yes	No	Partial	Partial	Yes
Macro analysis	Full	Partial	Weak	Weak	None
Data-flow taint	Yes	None	None	None	None
Deterministic	FP	Opaque	No	No	Yes
supp.					
Policy guardrails	Yes	None	Probes	Probes	None

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